

# Sample Test Report

Sample ID: SIVX-2025-00123

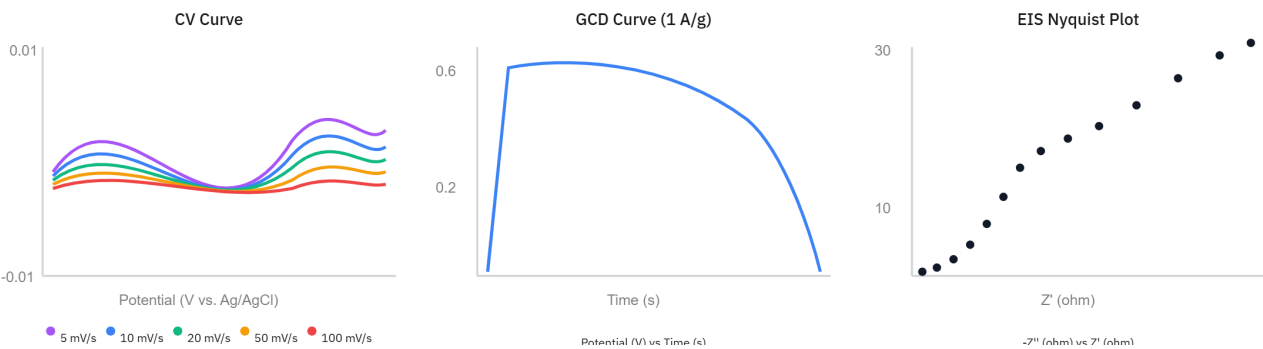
## Sample Details

|                    |                               |
|--------------------|-------------------------------|
| Material / Project | MnO2 Supercapacitor Electrode |
| Electrode Type     | Supercapacitor Electrode      |
| Substrate          | Nickel Foam (NF)              |
| Binder             | PVDF (10 wt%)                 |
| Total Mass Loading | ~2.1 mg/cm <sup>2</sup>       |
| Active Material    | MnO2                          |
| Test(s) Performed  | CV, GCD, EIS                  |

## Experimental Conditions

|                       |                         |
|-----------------------|-------------------------|
| Electrolyte           | 1 M KOH                 |
| Cell Type             | Two-electrode           |
| Potential Window      | -1.0 to 0 V             |
| Scan Rate (CV)        | 5, 10, 20, 50, 100 mV/s |
| Current Density (GCD) | 1, 2, 5, 10 A/g         |
| EIS Frequency Range   | 100 kHz - 0.01 Hz       |
| Temperature           | 25 ± 2 °C               |
| Instrument            | Biologic VSP            |
| Tested By             | SivionX Team            |
| Test Date             | 20 May 2025             |

## Report Preview (CV / GCD / EIS)



## Calculated Results

| Parameter                     | Value | Unit  | Method   |
|-------------------------------|-------|-------|----------|
| Specific Capacitance (1 A/g)  | 286.4 | F/g   | From GCD |
| Specific Capacitance (5 A/g)  | 232.7 | F/g   | From GCD |
| Specific Capacitance (10 A/g) | 189.3 | F/g   | From GCD |
| Energy Density                | 19.8  | Wh/kg | From GCD |

|                                  |       |      |          |
|----------------------------------|-------|------|----------|
| Power Density                    | 800   | W/kg | From GCD |
| IR Drop                          | 0.018 | V    | From GCD |
| Coulombic Efficiency             | 98.2  | %    | From GCD |
| Rs (Solution Resistance)         | 1.12  | ©    | From EIS |
| Rct (Charge Transfer Resistance) | 5.48  | ©    | From EIS |

### Interpretation

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- The CV curves exhibit quasi-rectangular shape with increasing current response, indicating good capacitive behavior and fast charge transport.
- The specific capacitance shows good retention with increasing current density.
- Low IR drop and small Rct value suggest low internal resistance and efficient ion diffusion.
- Overall, the electrode shows excellent supercapacitive performance.

### SivionX Remarks

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Well-fabricated electrode with high capacitance and low resistance.  
Optimization of mass loading can further improve energy density.